US

# Updesh Singh

**OPENING · THE PROBLEM**

WHAT TO SAY

Hello sir. Myself **Updesh Singh**, and we are **Team Deviar 3**. We are presenting our project **Samarth Sentinel**, built for problem statement DEV-104 from Samarth Life Management.

Sir, let me start with the problem as Samarth themselves describe it. They support **40,000 seniors across 350 cities**. Each Care Coordinator looks after many families, and can realistically call about **ten people a day**. Today they choose those ten from memory and old call notes.

So we asked ourselves one question. Is the real problem *predicting illness*? We decided it is not. The real problem is **"which ten people do I call this morning, and what do I say when they pick up?"**

That is a ranking problem under a hard daily limit. And sir, this one decision changed everything we built afterwards.

**150M+**
Indians aged 60+, many living alone

**~10/day**
what one coordinator can actually act on

**Days–weeks**
how early decline can be spotted

WHY THIS FRAMING MATTERS (SAY THIS SLOWLY)

Because we treat it as ranking under a budget, we measure ourselves on **precision at the top of the list** — not on accuracy, not on AUC. A model that ranks beautifully but floods the queue with 50 alerts is useless to the person holding the phone.

Decline in seniors is quiet. A missed medicine. Poorer sleep. Eating less. A fall nobody mentioned. Each signal is small on its own. Our system's job is to notice the *pattern*, days or weeks before it becomes an emergency — and then hand it to a human.

HAND OVER TO ANAMIKA

"Sir, that is the problem we are solving. Now my teammate **Anamika** will explain what we actually built and what makes our approach different."

AN

# Anamika

WHAT WE BUILT · WHAT IS NEW

WHAT TO SAY

Good morning sir. I am **Anamika**. Let me show you what we built and what is genuinely new in it.

Our system gives every senior a **daily risk score**, built from four layers working together — simple clinical rules, the senior's *own* personal baseline, a machine learning ranker, and an anomaly detector.

Sir, the second layer is the important one. We do not compare a senior to the general population. A blood pressure of 150 may be completely normal for one person and an emergency change for another. So we compare **each senior only against their own normal range**, calculated from their own past readings.

THREE THINGS THAT MAKE OUR APPROACH DIFFERENT

- **We show what we held back.** Our list is capped at ten. The seniors we scored but did not alert on are shown greyed out, right below the line. A coordinator who cannot see what was withheld has no reason to trust the list.
- **Silence is treated as evidence.** If a senior does not answer the daily call, we do not leave it blank — an unanswered call is itself a warning sign, and it is scored.
- **Every alert carries a sentence, not a number.** Instead of "risk 0.64", the coordinator reads "under 57% of doses logged this week" and "call this morning and go through the evening routine".

UNDERSTANDING REAL CONVERSATIONS

Sir, Indian call notes are not clean English. They are mixed — Hindi and English together. A coordinator writes "thoda dard hai" or "neend nahi aayi".

We built an extractor that reads this mixed language and pulls out **eight types of health cue** — pain, confusion, poor sleep, low appetite, loneliness, dizziness, breathlessness, and a fall.

And it understands **negation**. If the note says "gire nahi" — meaning *did not* fall — we record that it was asked, and we raise no alert. Getting that backwards would invent a fall that never happened.

HAND OVER TO RIDDHIMA

"Sir, now **Riddhima** will take you through our architecture and the technology we used."

RI

# Riddhima

ARCHITECTURE · TECH STACK

WHAT TO SAY

Good morning sir. I am **Riddhima**, and I will explain how this is actually built.

Sir, this is not a slide deck — it is a **working, deployed product**. You can open it right now on your phone.

The flow is simple. We generate a synthetic cohort of **200 seniors over 90 days**. That data goes through a feature layer, then the four scoring layers, then a ranking step that applies the daily cap. The output is alerts with reasons, stored in PostgreSQL. A Next.js web app reads it and shows four different screens for four different people.

**4**
separate user views, one database

**₹0**
runs on free cloud tiers, CPU only

**18,000**
senior-days scored and explained

PRIVACY IS IN THE DATABASE, NOT IN OUR CODE

Sir, this is the part I want to stress. A family member cannot see blood pressure, sugar, or call notes. But not because we wrote an `if` statement that hides them.

There is **no database permission granting them access at all**. So even if our website had a bug, the database returns zero rows. We tested this signing in as each role — family, operations, and a logged-out visitor. All blocked.

And when we use an AI service for the weekly family summary, **we never send the senior's real name**. We send a placeholder and put the name back on our own machine afterwards.

TECH STACK

**Next.js 15** · React 19 | **TypeScript** | **Supabase** · PostgreSQL | Row-Level Security | Supabase Auth + Google

**Python 3.11** | **scikit-learn** · Gradient Boosting | Isolation Forest | **Groq** · Whisper speech-to-text

Groq LLM · summaries | Web Speech API · text-to-speech | **Vercel** · deployed live

HAND OVER TO BHANU

"Sir, finally **Bhanu** will share our results and the impact this can have."

BH

# Bhanu

RESULTS · IMPACT · CLOSING

### WHAT TO SAY

Good morning sir. I am **Bhanu**, and I will close with our results and who this actually helps.

Sir, we planted **24 hidden decline cases** in our data. Our system caught **all 24** — and caught every one of them **before the day it became obvious**. On average, **22 days early**.

On precision, I want to be very honest with you. Our number is **0.169**. But sir, please look at the ceiling. With 24 real cases spread across 90 days and only ten slots per day, the *mathematically highest possible* precision is **0.192**. Most days genuinely have fewer than ten declining seniors. So we are reaching **88% of what is even achievable**.

**24 / 24**
planted declines caught, all before the obvious day

**22 days**
median early warning

**88%**
of the mathematically achievable precision

### WHO THIS HELPS

**The coordinator** gets a short, ordered list with a reason and an action — something to act on tomorrow morning, not a dashboard to study.

**The senior** needs no app and no smartphone. Six short questions on a normal phone call, answered by pressing 1, 2 or 3, in Hindi or English — on the handset they already own.

**The family** gets a warm weekly note in plain language, no readings and no scores. They can also write back to the coordinator — because a daughter knows things no sensor records.

### BEING HONEST ABOUT LIMITS — SAY THIS, IT MATTERS

Sir, one thing did **not** work. We built the feedback loop where coordinators mark alerts useful or not. We measured it carefully, and it improved precision by almost nothing. We are reporting that honestly instead of hiding it. We also do not detect falls from sensors — only when a senior mentions it.

### CLOSING LINE

Sir, Samarth Sentinel never diagnoses. It notices a change, explains it in plain words, and puts it in front of a human who can pick up the phone. **Thank you, sir.**